

Inferring critical transitions from timeseries

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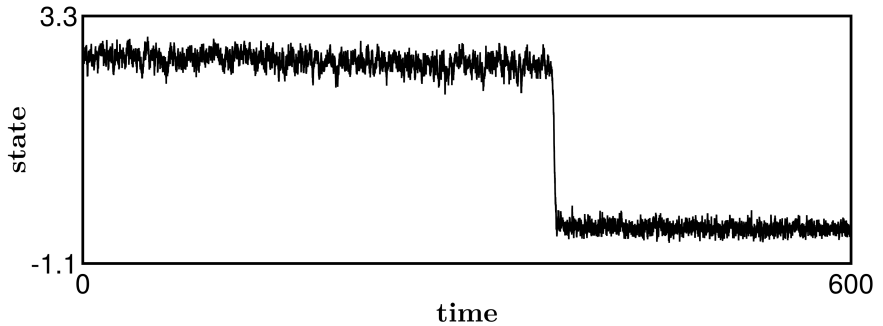
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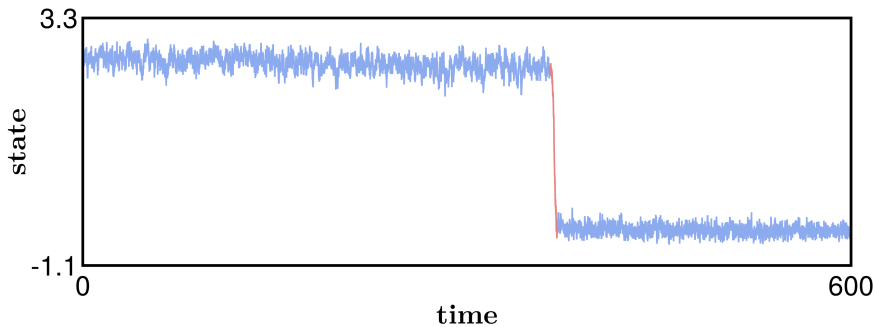
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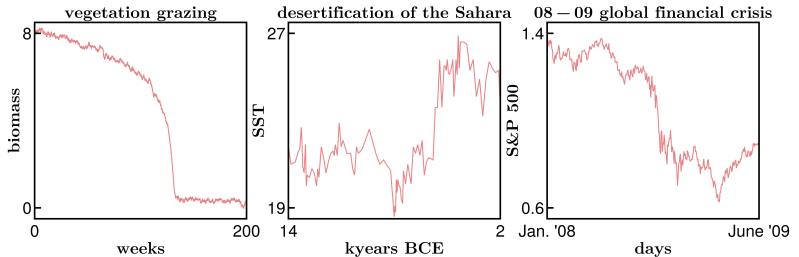
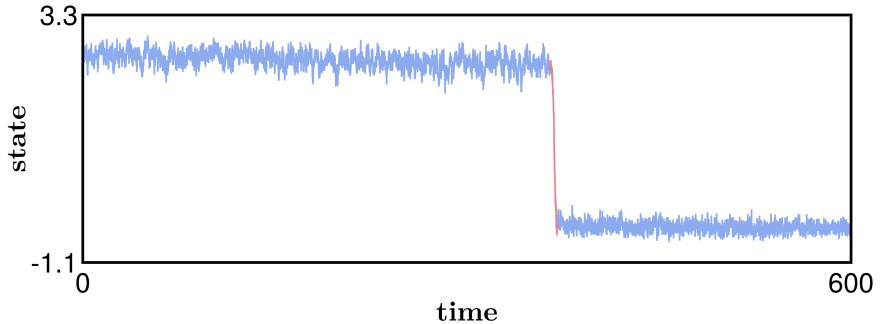
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Dynamical systems that tip

Consider n -dimensional Itô processes with additive diffusion $\sigma = \sqrt{2D} > 0$ and non-autonomous drift

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The law of evolution of the parameter is a **parametrised family of shifts** in time connecting the past-limit (i.e. $\mu(t) \rightarrow \mu_-$, $t \rightarrow -\infty$) to the future-limit system (i.e. $\mu(t) \rightarrow \mu_+$, $t \rightarrow +\infty$).

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If at any point $\mu(t)$ a QSE is created, destroyed or it changes its stability then a **bifurcation** has occurred.

Tipping mechanisms

We want to differentiate between bifurcation-induced (B-tipping) from rate-induced (R-tipping) critical transitions.

The former type (usually) happens when the parameter shift is too abrupt.

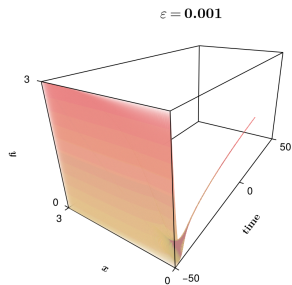
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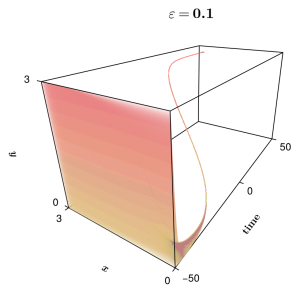
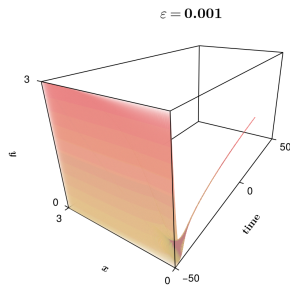


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$$X_t = X_0 + \int_0^t (-\mu - X_s^2) ds + \int_0^t \sqrt{2D} dW_s.$$

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Assuming we start the process at fixed $X_0 = \sqrt{-\mu} + \delta$ ($\delta \ll 1$), a solution in a neighbour of $X_t = \sqrt{-\mu}$ is well approximated by the solution of the linear Ornstein-Uhlenbeck process (OUP)

$$X_t = \theta \int_0^t (\sqrt{-\mu} - X_s) ds + \sqrt{2D} \int_0^t dW_s,$$

which admits a stationary probability density with $\theta = -2\sqrt{-\mu}$ and covariance

$$\text{Cov}[X_s, X_t] = \frac{D}{\theta} e^{-\theta(t-s)}, \quad s < t.$$

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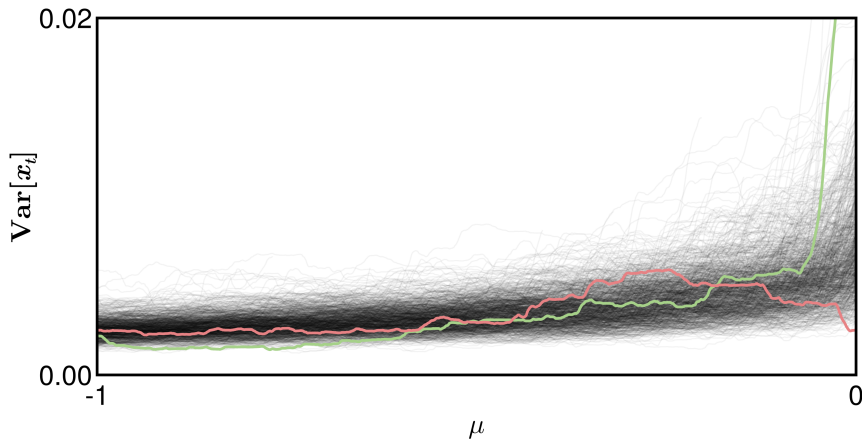
$$\text{Cov}[X_s, X_t] = \frac{D}{\theta} e^{-\theta(t-s)}, \quad s < t.$$

If we now consider instantaneous time increments (i.e. $t \rightarrow s$) then

$$\text{Cov}[X_s, X_t] \rightarrow \text{Var}[X_t] = \frac{D}{\theta} \rightarrow \begin{cases} 0, & \mu \rightarrow -\infty, \\ \infty, & \mu \rightarrow 0. \end{cases}$$

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$$f(X_t; \mu) = -\mu - X_t^2 \implies \text{Var}[X_t] = \frac{D}{\theta} \rightarrow \begin{cases} 0, & \mu \rightarrow -\infty, \\ \infty, & \mu \rightarrow 0. \end{cases}$$



Entering large deviation theory

Theorem (Freidlin-Wentzell)

The probability of the sample paths $X^D = \{X_t\}_{t=0}^T$ of $dX_t = f(X_t)dt + \sqrt{2D}dW_t$ to be located in a cylinder of radius $\delta > 0$ around any (absolutely continuous) deterministic path ϕ decays as

$$P[\phi] \asymp \exp\left(-\frac{S[\phi]}{2D}\right) \quad \left(\text{or} \quad \lim_{D \rightarrow 0} \log P\left(\sup_{t \in [0, T]} |X_t - \phi(X_0, t)| < \delta\right) = -S[\phi]\right),$$

where the cost of following such path is given by the rate functional

$$S[\phi] = \int_0^T L(\dot{\phi}, \phi, t) dt = \frac{1}{2} \int_0^T \left(\dot{\phi} - f(\phi)\right)^2 dt,$$

In the singular limit $\varepsilon \rightarrow 0$ the most likely path $\phi(X_0, t) \in H^1(\mathbb{R}^d, [0, T])$ is the minimiser of $S[\phi]$ and it satisfies the Euler-Lagrange equations (ELEs)

$$\frac{d}{dt} \left(\frac{\partial}{\partial \dot{\phi}} L(\dot{\phi}, \phi, t) \right) - \frac{\partial}{\partial \phi} L(\dot{\phi}, \phi, t) = 0.$$

Treating tipping points as rare events

Corollary

Consider a 1-dimensional, conservative vector field f (i.e. $\exists V$ s.t. $f = -\nabla V$) with a unique, stable fixed point x_0 . Then the probability for hitting a target x scales as $\exp(V(x_0) - V(x))$.

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Consider a vector field $f(x; \mu)$ with equilibria $x_{\text{stb}, \text{uns}}$.

If the vector field is conservative then its potential $V(x; \mu)$ has a local minima at $a \equiv x_{\text{stb}}$ and a local maxima at $b \equiv x_{\text{uns}}$.

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Approaching a saddle-node bifurcation ($\mu \rightarrow \mu_{\text{bif}}$) we have $|x_{\text{stb}} - x_{\text{uns}}| = |a - b| \rightarrow 0$, meaning that

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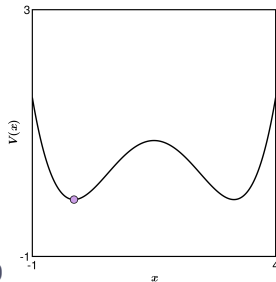
We can use the LDP as an EWS!

$$\exp(-\Delta V) = \begin{cases} 0, & \text{if } |\mu - \mu_{\text{bif}}| \rightarrow \infty, \\ 1, & \text{if } |\mu - \mu_{\text{bif}}| \rightarrow 0. \end{cases}$$

Transition probabilities

Given a 1-dimensional conservative system $dX_t = -V'(X_t; \mu)dt + \sqrt{2D}dW_t$ the transition probability $p(X_t, t|X_0, 0)$ satisfies the Fokker-Plank equation

$$\partial_t p = \mathcal{L}p := (V'p)' + Dp''.$$



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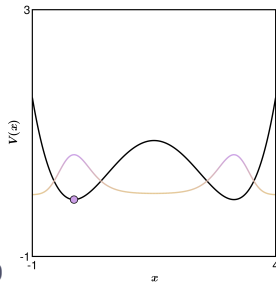
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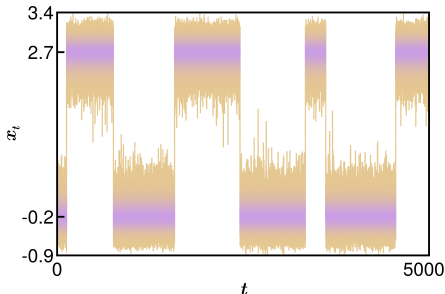
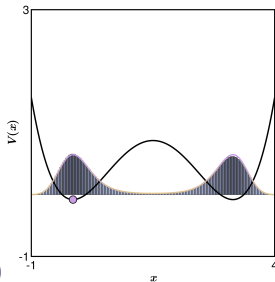
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Functional framework

An observed sample path can be thought of as a sequence $\{X_n\}_{n=1}^N \subset I \subseteq \mathbb{R}$ of $N > 0$ i.i.d. random variables sampled from $p(x) \in H := \{f \in L^2(\mathbb{R}) : \int_{\mathbb{R}} f(x)dx = 1\}$.

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Fixing $h > 0$ identifies a unique partition I_h of I into $0 < M < N$ homogeneous (i.e. fixed width), disjoint subintervals

$$I_h = \cup_{k=1}^M B_k, \quad B_k = [\min\{X_t\} + (k-1)h, \min\{X_t\} + kh).$$

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Proposition

There exists a unique minimiser of

$$p_h := \operatorname{argmin}_{f \in \mathcal{H}_h} \|f - p\|_H,$$

and it coincides with the bin-averaged true density, i.e. $c_k = \frac{1}{h} \int_{B_k} p(x)dx =: p_k$.

Maximum likelihood estimation

The (empirical) histogram associated to the sample $\{X_n\}_{n=1}^N$

$$\hat{p}_h(x) = \sum_{k=1}^M \hat{p}_k \mathbf{1}_{B_k} = \frac{1}{Nh} \sum_{k=1}^M \mathbf{1}_{B_k}(x) \sum_{n=1}^N \mathbf{1}_{B_k}(X_n),$$

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$$\sqrt{N}(\hat{p}_k - p_k) \xrightarrow{d} \mathcal{N}\left(0, \frac{\pi_k(1 - \pi_k)}{h^2}\right), \quad k = 1, \dots, M,$$

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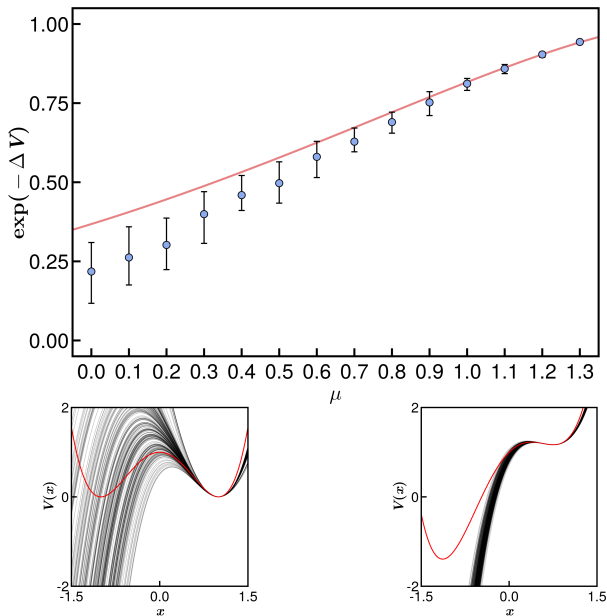
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Given the (empirical) histogram $\{\hat{x}_k, \hat{p}_k\}$ over the bins' center points $\hat{x}_k$, find

$$\theta^* = (c_j)_{j=0,\dots,3} = \operatorname{argmin}_{\theta \in \mathbb{R}^4} \sqrt{\sum_{k=1}^M (\exp(-U(\hat{x}_k; \theta)) - \hat{p}_k)^2}.$$

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Thank you!

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